General Relativity II / Cosmology

Fall 2021

Homework Assignment No. 1

Problem's in Carroll's Book

Chapter 5 Problems: 1, 2(a)-(d), 3, 4

for 4(b), in sketches
set angular momentum $L = 0$
and consider separate cases $\Lambda > 0, \Lambda = 0$, and $\Lambda < 0$.

Problem Ex1.

For the Schwarzschild metric,

(a) Show that the Kretschmann scalar $R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}$ is finite at $r = 2GM$ and diverges at $r = 0$.

Hint: to do this, first calculate the terms: $R^{\alpha\beta}{}_{\alpha\beta}$. (no sum over repeated indices)

(b) What are the values of t and r in the spherical polar coordinate system at the central point of the Kruskal diagram?
What are the values of all non-zero components of $R^{\alpha\beta}{}_{\alpha\beta}$ ($\alpha, \beta = T, R, \theta, \phi$) at this point?

Problem Ex2.

For the Schwarzschild metric, consider a tachyon of mass M following a space-like trajectory.

The relativistic energy of the tachyon at infinity is E .

Show that there is a smallest value of r, the radial coordinate, reachable by the tachyon, and find its value.

Problem Ex3.

The dwarf Planet **Pluto** and **Halley's Comet** have orbits with the following characteristics:





	Pluto ♇	**Halley's**
perihelion:	29.658 A.U.	0.586 A.U.
aphelion:	49.305 A.U.	35.082 A.U.
eccentricity:	0.2488	0.96712
period:	248.00 years	75.32 years

Calculate, in units of arc seconds per century, the perihelion precession rate *due to general relativity* (alone) for each of these bodies.

Problem Ex4.

For the supermassive black hole at the center of our galaxy, assume $M = 10^6 M_\odot$.

Also assume $a = 0$ (not Kerr) and $Q,P = 0$ (not RN).

(a) What is this black hole's Schwarzschild radius in A.U. ?

(b) A super-advanced orbiting telescope is observing the black hole from the smallest possible stable circular orbit. Once per orbit, the telescope sends a burst of data to the planet Bi, home of a very advanced civilization and far, far away from the galactic center. According to the telescope's internal clock, what is the time interval between transmission bursts?
And, according to the inhabitants of Planet Bi, what is the time interval between reception of the sequential bursts?

EXTRA Bonus Problem Ex5.



Kip Thorne
Nobel Prize in 2017 for LIGO

In the 2014 movie ***Interstellar*** , on which Kip Thorne was a consultant and executive producer, there is a planet orbiting a massive black hole, $M = 10^8 M_\odot$, such that 1 hour on the planet corresponds to 7 years on Earth. Assume that the black hole is described by the *Kerr metric* and that the planet is in geodesic motion in the innermost stable circular orbit. What is the orbital radius of the planet and what is the angular momentum of the black hole?



Note: I do NOT have a complete solution to this problem !!!
I put it out to you as an **OPTIONAL** challenge to the students !!!!!

ROTATING BLACK HOLES: LOCALLY NONROTATING FRAMES, ENERGY EXTRACTION, AND SCALAR SYNCHROTRON RADIATION*

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Circular orbits do not exist for all values of r. The denominator of equations (2.12) and (2.13) is real only if

$$r^{3/2} - 3Mr^{1/2} \pm 2aM^{1/2} \geq 0 . \tag{2.17}$$

The limiting case of equality gives an orbit with infinite energy per unit rest mass, i.e., a photon orbit. This photon orbit is the innermost boundary of the circular orbits for particles; it occurs at the root of (2.17),

$$r = r_{\rm ph} \equiv 2M\{1 + \cos[\tfrac{2}{3}\cos^{-1}(\mp a/M)]\} . \tag{2.18}$$

For $a = 0$, $r_{\rm ph} = 3M$, while for $a = M$, $r_{\rm ph} = M$ (direct) or $4M$ (retrograde).

For $r > r_{\rm ph}$ not all circular orbits are bound. An unbound circular orbit is one with $E/\mu > 1$. Given an infinitesimal outward perturbation, a particle in such an orbit will escape to infinity on an asymptotically hyperbolic trajectory. The unbound circular orbits are circular in geometry but hyperbolic in energetics, and they are all unstable. Bound circular orbits exist for $r > r_{\rm mb}$, where $r_{\rm mb}$ is the radius of the marginally bound ("parabolic") circular orbit with $E/\mu = 1$,

$$r_{\rm mb} = 2M \mp a + 2M^{1/2}(M \mp a)^{1/2} . \tag{2.19}$$

Note also that $r_{\rm mb}$ is the minimum perihelion of all parabolic ($E/\mu = 1$) orbits. In astrophysical problems, particle infall from infinity is very nearly parabolic, since the velocities of matter at infinity satisfy $v \ll c$. Any parabolic trajectory which penetrates to $r < r_{\rm mb}$ must plunge directly into the black hole. For $a = 0$, $r_{\rm mb} = 4M$; for $a = M$, $r_{\rm mb} = M$ (direct) or $5.83M$ (retrograde).

Even the bound circular orbits are not all stable. Stability requires that $V_r''(r) \leq 0$, which yields the three equivalent conditions

$$1 - (E/\mu)^2 \geq \tfrac{2}{3}(M/r) ,$$

$$r^2 - 6Mr \pm 8aM^{1/2}r^{1/2} - 3a^2 \geq 0 ,$$

or

$$r \geq r_{\rm ms} , \tag{2.20}$$

where $r_{\rm ms}$ is the radius of the marginally stable orbit,

$$r_{\rm ms} = M\{3 + Z_2 \mp [(3 - Z_1)(3 + Z_1 + 2Z_2)]^{1/2}\} ,$$
$$Z_1 \equiv 1 + (1 - a^2/M^2)^{1/3}[(1 + a/M)^{1/3} + (1 - a/M)^{1/3}] ,$$
$$Z_2 \equiv (3a^2/M^2 + Z_1^2)^{1/2} . \tag{2.21}$$